

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Operating Systems

COURSE CODE:CSA0404

COURSE OUTCOME COVERED

CO4: Optimize various file allocation strategies and disk scheduling algorithms to identify optimal methods for enhancing system performance.

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

INDUSTRY PROBLEM SOLVING TASK

Code of Conduct:

I BP VENKATA UDAY KUMAR 192411203 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

UDAY KUMAR

Signature of the student:

Annexure A

INDUSTRY PROBLEM SOLVING TASK

S.No	Industry Problem Solving Task Questions	Marks
1	A cloud storage company is facing performance issues due to frequent file access delays. Analyze whether contiguous, linked, or indexed allocation would be most suitable for improving access speed and justify your choice with reasons.	5
2	A data center experiences high disk head movement while processing user requests. Suggest the most appropriate disk scheduling algorithm (FCFS, SSTF, SCAN) to reduce seektime and explain how it improves performance in this scenario.	5
3	An operating system uses a bitmap for free space management but struggles with slow allocation when the disk size increases. Propose an optimization technique to improve the efficiency of bitmap-based allocation and justify your solution.	5
4	A real-time application requires consistent and predictable disk access times. Evaluate whether SCAN or C-SCAN would be more suitable and explain your reasoning based on system requirements.	5
5	A company wants to design a hybrid storage system combining file allocation and disk scheduling techniques. Suggest a suitable combination (allocation + scheduling) for handling large-scale data access efficiently and justify your choice based on performance factors.	5

1. Cloud Storage – File Allocation

Recommended solution: Contiguous Allocation

A cloud storage company needs fast access because frequently requested files can create delays. Contiguous allocation is the most suitable when high access speed is the main requirement.

- In contiguous allocation, all blocks of a file are stored continuously on the disk.
- It provides very fast sequential and direct access because the disk head does not need to move between widely separated blocks.
- It reduces seek time and improves overall file-access performance.
- It also provides good locality, which is useful for frequently accessed files.
- However, it may suffer from external fragmentation and difficulty expanding files.
- In a cloud environment, these limitations can be reduced using storage management and file-size prediction techniques.

Performance justification: Compared with linked allocation, contiguous allocation requires fewer block-pointer traversals. Compared with indexed allocation, it avoids additional index-block access.

Conclusion: Contiguous allocation is preferred because it provides high-speed file access and lower access overhead, which directly addresses the company's performance problem.

2. Data Center – Disk Scheduling

Recommended solution: SCAN

The data center experiences high disk-head movement because many user requests may arrive at different disk locations. SCAN scheduling is suitable because it reduces unnecessary movement while maintaining fairness.

- SCAN is also called the Elevator Algorithm.
- The disk head moves in one direction and services all requests encountered along the way.
- When it reaches the end, it reverses direction and continues servicing requests.
- This reduces unnecessary seek movements compared with FCFS.
- Unlike SSTF, SCAN provides better fairness because requests are less likely to suffer from starvation.
- It performs well when there are many requests distributed across the disk.

Performance justification: SCAN provides a better balance between seek time, throughput, and fairness than FCFS and SSTF in a heavily loaded data-center environment.

Conclusion: SCAN is the most appropriate choice because it reduces average head movement while providing fair service to multiple requests.

3. Bitmap – Free Space Management

Recommended solution: Hierarchical Bitmap

A bitmap becomes slower to search as disk capacity increases because the operating system may need to scan a very large number of bits to locate free blocks.

A hierarchical bitmap can improve the allocation process.

- Divide the large bitmap into smaller groups.
- Create a higher-level bitmap that indicates whether each group contains free blocks.
- First search the higher-level bitmap.
- Once a suitable group is identified, search only that group's bitmap.
- This significantly reduces the number of bitmap entries that must be examined.
- The method is particularly effective for large-capacity storage systems.

Example:

Instead of searching the entire bitmap:

0000000000000000000010000...

The system first identifies the group containing free space and then searches only that group.

Performance justification: The optimization reduces the free-block search time and allocation overhead, making bitmap-based management more scalable.

Conclusion: A hierarchical bitmap is a practical optimization because it improves allocation speed, scalability, and storage-management efficiency.

4. Real-Time Application – SCAN vs C-SCAN

Recommended solution: C-SCAN

A real-time application requires consistent and predictable disk-access time. Therefore, C-SCAN is more suitable than SCAN.

- C-SCAN services requests while moving in one direction only.
- When the head reaches the end, it returns to the beginning without servicing requests during the return trip.
- Requests are therefore handled in a more uniform manner.
- SCAN changes direction and may provide different waiting times depending on the request's location.
- C-SCAN provides a more predictable response-time distribution.
- Predictability is important in real-time systems where meeting timing requirements is critical.

Performance justification: C-SCAN may sometimes involve more physical movement than SCAN, but it provides more uniform waiting times, which is more important for real-time applications.

Conclusion: C-SCAN is preferred because it provides consistent and predictable access times, satisfying the main requirement of the real-time system.

5. Hybrid Storage System

Recommended combination: Contiguous Allocation + SCAN

For a large-scale storage system, a combination of an efficient file allocation technique and disk scheduling algorithm is required.

File Allocation – Contiguous

- Stores file blocks close together.
- Provides fast sequential and direct access.
- Reduces unnecessary disk-head movement during file access.
- Improves read performance for frequently accessed large files.

Disk Scheduling – SCAN

- Services multiple requests while moving across the disk.
- Reduces unnecessary seek operations.
- Provides better fairness than FCFS.
- Works effectively under heavy workloads with many requests.

Performance analysis:

The combination can improve:

- **Access speed**
- **Seek-time efficiency**
- **Throughput**
- **Fairness**
- **Overall disk utilization**

A simulation using different request sequences and file sizes can be used to compare average seek time, total head movement, and response time.

Conclusion: Contiguous Allocation + SCAN is a suitable hybrid solution because it combines fast file access with efficient and fair disk scheduling, making it practical for large-scale data access.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis